

MODEL BASED INFERENCE FOR NETWORK STRUCTURED DATA, WITH APPLICATION TO BRAIN FUNCTIONAL CONNECTIVITY STUDY

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ABSTRACT

This paper is motivated by brain connectivity studies in translational neuroscience, where there is an interest in understanding how functional connectivity (network structure) in brains develops with age, and interacts with covariates such as behavioral measures, cognitive function and mental disorder status. We propose a covariate-incorporated-semiparametric blockmodel to improve estimation and inference of subject level covariate effects on network structured data. Simulation results show that our proposed method has small estimation error and good statistical power. We applied our method to a resting state fMRI data set and found that the connectivity in basal ganglia shows a developmental local-to-distributed trend through adolescence and the trend is positively associated with IQ values.

KEY WORDS: blockmodel, brain connectivity, network, resting state fMRI, semiparametric, random effects.

1 Introduction

Brain connectivity is assessed through the characterization of white matter structural connectivity, and maps of the brain’s functional interactions; the latter is usually described as a connectivity matrix, whose elements index the degree of interaction between pairs of local activities. In recent years, non-invasive neuroimaging techniques such as fMRI and MEG/EEG exhibit vastly improved spatial and temporal resolution, which makes it feasible to quantify different aspects of functional connectivity in the brain for a sample of subjects (Astolfi et al. 2007; Rogers et al. 2007; Fornito et al. 2012; Van Essen et al. 2012). For each subject i , functional connectivity is characterized by a matrix $y_{i,jl}$, $i = 1, \dots, n; j, l = 1, \dots, m$. The measure of $y_{i,jl}$ can be application-specific, for example, Pearson correlation of time course activities between voxels j and l from fMRI data, and frequency specific phase coherence across trials from MEG/EEG data.

One of the main interests of functional connectivity study in translational neuroscience is to investigate how functional connectivity (network structure) develops with age (Hwang et al. 2012; Dosenbach et al. 2010), and interacts with covaraites such as behavioral measures, cognitive function and mental disorder status. The inference for brain connectivity network is a very hard problem and a commonly adopted routine in neuroscience literature is to first study the brain connectivity for each subject and get a few summarizing quantities $\tilde{y}_i$ by graph theory or simply averaging over pairs of nodes, and followed by an add-on step to regress $\tilde{y}_i$ on covarites X_i . However, the summary statistics obtained for each subject are likely to contain a large subject level random fluctuation, and the subject level covariate effects can hardly be detected with a relatively small sample size n . Note that one of the prominent features for brain connectivity data computed from neuroimaging is that we usually have a relatively small number of subjects n but a very large number m^2 pairs of nodes. The structure of brain connectivity data $y_{i,jl}$ is similar to clustered data or longitudinal data

in some sense, where we have independent subjects and correlated measurements within each subject. Therefore to better utilize the data, an integrated modeling of covariate effects on $y_{i,jl}$ is highly desired.

In this paper, we propose a covariate-incorporated-semiparametric blockmodel. We first assume that all the nodes can be partitioned into r disjoint regions of interest or equivalently each node j has an $r \times 1$ label vector $R_j = (0, 0, 1, \dots, 0)^T$, where only one entry equals 1 and it indicates the region that node j belongs to. For all m nodes, we have a label matrix $\mathbf{R} = (R_1, R_2, \dots, R_m)^T$, with dimension $m \times r$. We also note that in the brain connectivity study, all the nodes are naturally embedded in a three-dimensional brain space, and recorded activities in adjacent nodes usually are correlated due to technique reasons. Here we briefly describe the model for $y_{i,jl}$ as follows (rigorous definitions can be found in Section 2.1).

$$y_{i,jl} = R_j' C(X_i) R_l + f(d_{jl}) + g_i(d_{jl}) + \epsilon_{i,jl}, \quad (1)$$

where $C(X_i)$ is an $r \times r$ matrix with a linear form in X_i (will be specified in Section 2.1). Here d_{jl} is the Euclidean distance between voxel j and voxel l , and $f(d_{jl})$ is a smooth function of d_{jl} to account for the spurious connectivity in adjacent nodes. The random functions $g_i(d_{jl})$ are independent realizations of a mean zero random process, accounting for within subject correlations. Before we discuss our data application in detail, one can view Figure (1) as a simple illustration of the model.

Model (1) implies that, given the covariate X_i , for j and j' in the same brain region, as well as l and l' in the same brain region, the expected functional connectivity, after removing spatial contamination, are the same for $j \leftrightarrow l$ and $j' \leftrightarrow l'$. This can be viewed as a dimension reduction of the connectivity data and compress the redundant information presents at the voxel scale. On one hand, model (1) shares some common features with the widely known stochastic blockmodel used in technological and social networks, which is characterized by the fact that each node belongs to one of multiple blocks and the probability of a relationship between two nodes depends only on the memberships of the two nodes. On the other hand,

our model addressed unique features and challenges present in the estimation and inference for brain connectivity network. Special features include multiple realizations on a sample of subjects with subject level covariates and individual fluctuations, a natural spatial embedding in a three-dimensional brain space, and continuous connectivity measures contaminated with non-negligible errors. The semiparametric model described above is particularly appealing in our application, since the effects of X are of major interest, while $f(d_{jl})$ is a nuisance part.

To estimate the model components, we first consider model (1) for a given brain partition $\mathbf{R}$. Then the first part $R_j' C(X_i) R_l$ is basically a parametric part and $f(d_{jl})$, $g_i(d_{jl})$ are nonparametric parts. Efficient estimation for semiparametric models has been extensively studied. For independent data, one of the most popular and efficient methods is the profile kernel methods, where one uses the kernel method to estimate the nonparametric functions and the profile likelihood based method to estimate the parametric part (Hastie & Tibshirani 1990; Severini & Staniswalis 1994; Bickel et al. 1993), and recently Wang et al. (2011) proposed a difference based approach for a partial linear model which is asymptotically efficient and simple to implement. For clustered data or longitudinal data, how to effectively take into account within-cluster correlation has been a major concern, and different methods have been proposed to try to get an efficient estimate for the linear coefficients of the main covariates X (Zeger & Diggle 1994; Lin & Carroll 2001b,a; Fan & Li 2004; Wang et al. 2005; He et al. 2005; Huang et al. 2007). However, these methods are mainly for longitudinal studies or clustered data with bounded cluster sizes, where one could potentially get an efficient estimate by specifying the covariance structure appropriately. In our application, an intrinsic difference is that within each subject we observe a connectivity matrix and the number of pairs of nodes m^2 is much larger than the sample size n . In addition to a nonparametric fixed part f , we also have a nonparametric random part g_i , whose covariance structure could be very complex and hard to specify. In this paper, we extend the difference based approach

(Wang et al. 2011) from independent data to the new clustered data structure. In Section 2.2, we propose a very simple and effective algorithm to estimate all the model components, including error variance, where one does not need to specify any within subject covariance structure. The only assumption needed for f and g_i is smoothness.

In brain connectivity analysis, the partition of functionally distinct regions of interest can be informed from previous knowledge, determined by experimental purpose or obtained through a data driven partition procedure. We will discuss a data-driven partition algorithm based on the proposed model in Section 2.3 and simulation results show that the proposed inference is not sensitive to moderate errors in the partition step. The data driven brain partition could be of independent interest (Craddock et al. 2012; Cohen et al. 2008; Beckmann et al. 2009). Several procedures have been proposed to fit stochastic blockmodels. Bickel & Chen (2009) showed that the maximizers of the likelihood based modularity are asymptotically consistent estimators of block partitions. Similar results has been obtained for degree-corrected blockmodels by Zhao et al. (2012). To obtain a data-driven partition based on the proposed semiparametric blockmodel, in Section 2.3, we propose to combine a backfitting algorithm (Hastie & Tibshirani 1986) with a modified labeling switch algorithm (Stephens 2000; Bickel & Chen 2009) to approximate the optimal solution to the likelihood based modularity.

The proposed method is a new framework and among the first that aims at improving estimation and inference of subject level covariate effects on network structured data. We applied our method to a resting state fMRI data set and found that the connectivity in basal ganglia shows a developmental local-to-distributed trend through adolescence and the trend is positively associated with IQ values. Although this paper is inspired and illustrated by the brain connectivity study, our proposed methods and theoretical results pertain to a general setting and can be used in a variety of situations where one needs to model the evolution of a network at different levels of covariates.

The paper is organized as follows. In Section 2, we introduce the proposed model and the estimating procedure. In Section 3, we apply the proposed method to an RS-fMRI data to investigate the development of the brain network. This is followed by two numerical studies in Section 4 to illustrate some aspects of the proposed method. Further discussions can be found in section 5. Some technical proofs are in the appendix.

2 Statistical modeling and estimation

2.1 Covariate Incorporated Semiparametric Blockmodel

We start with brain connectivity measures $y_{i,jl}$, where $i = 1, \dots, n$, is the index of subjects, $j, l = 1, \dots, m$ are the index of nodes in a connectivity study. In our application, nodes are usually small voxels or patches in the brain. Depending on the spatial resolution of the particular imaging techniques, m ranges from hundreds to thousands. Assume all the nodes can be partitioned into r disjoint regions of interest or equivalently each node j has an $r \times 1$ label vector $R_j = (0, 0, 1, \dots, 0)^T$, where only one entry equals 1 and it indicates the region node j belongs to. For all m nodes, we have $am \times r$ label matrix $\mathbf{R} = (R_1, R_2, \dots, R_m)^T$.

We use an $r \times r$ matrix C to denote the fixed effect part of the integrity of functional activities between each pairs of regions, and this matrix depends on subject level covarites X . More specifically $C_{kk'}(X)$ is the entry on k th row and k th column of the matrix $C(X)$, denoting connectivity level between region k and region k' . For each subject, given the subject level covariate X_i , subject level deviation g_i , and node labels $\mathbf{R}$. We have the model as in Eq. (1)

$$y_{i,jl} = R_j' C(X_i) R_l + f(d_{jl}) + g_i(d_{jl}) + \epsilon_{i,jl},$$

where $y_{i,jl}$ is the connectivity value between voxels j and l , d_{jl} is the Euclidean distance between note j and node l , $f(0) = 0$, $E(g_i(\cdot)) = 0$ for identifiability, and $\epsilon_{i,jl}$ are independent measurement errors with mean zero and variance σ_ϵ^2 .

Our main interest is how $C_{kk'}(X)$ depends on X , and we consider a linear relationship. For each pair (j, l) , define dummy variables Z as follows. For $1 \leq k \leq k' \leq r$, we have

$$Z_{jl, kk'} = \begin{cases} 1 & \text{for } k = k' = 1 \\ 1 & \text{for } j \text{ in region } k \text{ and } l \text{ in region } k' \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

Then,

$$y_{i, jl} = \sum_{1 \leq k \leq k' \leq r} (\alpha_{kk'} Z_{jl, kk'} + \beta_{kk'} X_i \times Z_{jl, kk'}) + f(d_{jl}) + g_i(d_{jl}) + \epsilon_{i, jl}, \quad (3)$$

where $\alpha_{kk'}$ and $\beta_{kk'}$ can be interpreted as relative incremental connectivity using the pair $1 \leftrightarrow 1$ as reference. Note that for multivariate covariates, $X_i = (X_{i1}, X_{i2}, \dots, X_{id})^T$, and $\beta_{kk'} = (\beta_{kk', 1}, \beta_{kk', 2}, \dots, \beta_{kk', d})$.

2.2 Difference Based Estimating Procedure

Before we discuss the estimation procedure, for notation simplicity, we transform the double index to single index by $p = (k - 1) \times r + k'$, and $q = (j - 1) \times m + l$. We can rewrite the model as

$$y_{i, q} = \sum_{p=1}^{r^2} (\alpha_p Z_{q, p} + \beta_p X_i \times Z_{q, p}) + f(d_q) + g_i(d_q) + \epsilon_{iq}, \quad (4)$$

where $i = 1, \dots, n$, $q = 1, \dots, m^2$. Here β_p with single index is the same as $\beta_{kk'}$ with double index, for $p = (k - 1) \times r + k'$.

Remark: For undirected connectivity, we will only use data in upper triangle, i.e., $q = 1, \dots, m(m + 1)/2$, $p = 1, \dots, r \times (r + 1)/2$. In the following presentation, we still use all data for notation simplicity.

Algorithm Part I: The difference procedure contains three main steps.

1. Within each subject, order the observations $y_{i, q}$ according to the values of d_q , and we always refer to the ordered data thereafter. Also we assume X has been centered, i.e., $\sum_{i=1}^n X_i = 0$.

2. Within each subject, we obtain the new response by the difference procedure

$$D_{i,q} = \frac{1}{\sqrt{2}}(Y_{i,2q} - Y_{i,2q-1}), \quad \text{for } q = 1, \dots, m^*,$$

where $m^* = \lfloor \frac{m^2}{2} \rfloor$. Then we consider

$$D_{i,q} = \sum_{p=2}^{r^2} \left(\alpha_p \tilde{Z}_{q,p} + \beta_p X_i \times \tilde{Z}_{q,p} \right) + \delta_{i,q} + \varepsilon_{i,q}, \quad (5)$$

where $\tilde{Z}_{q,p} = \frac{1}{\sqrt{2}}(Z_{2q,p} - Z_{2q-1,p})$, $\delta_{i,q} = \frac{1}{\sqrt{2}}(f(d_{2q}) + g_i(d_{2q}) - f(d_{2q-1}) - g_i(d_{2q-1}))$, and $\varepsilon_{i,q} = \frac{1}{\sqrt{2}}(\epsilon_{i,2q} - \epsilon_{i,2q-1})$. We note that α_1, β_1 are canceled out in the difference procedure.

Vectorize D , and put all covariates together in a matrix Q . We have

$$D = QB + E,$$

where D is $nm^* \times 1$, $B = (\alpha_2, \dots, \alpha_{r^2}, \beta_2, \dots, \beta_{r^2})^T$, with dimension $p \times 1$, Q is $nm^* \times p$, and E contains $\delta + \varepsilon$. At this point, we argue heuristically that δ is negligible comparing to ε , since f and g_i are smooth functions. We estimate B by simple least square methods.

$$\hat{B} = (Q'Q)^{-1}QD \quad (6)$$

3.

$$\hat{\sigma}_\epsilon^2 = \frac{1}{nm^* - \text{rank}(Q)}(D - Q\hat{B})^T(D - Q\hat{B})$$

Remark: In the application of brain functional connectivity study, $nm^* = n\lfloor m^2/2 \rfloor$ can be a huge number, and the matrix Q might exceed storage. However, the number of distinct rows in Q will not exceed $n^* = n(r^2 - r + 1)$. For computational convenience, we define a compressed matrix $\tilde{Q}$, whose j th line is the j th distinct value in Q , $j = 1, \dots, n^*$, without lost of generality, we assume

$$Q = (\underbrace{\tilde{Q}_1, \dots, \tilde{Q}_1}_{w_1}, \dots, \underbrace{\tilde{Q}_{n^*}, \dots, \tilde{Q}_{n^*}}_{w_{n^*}}).$$

Accordingly, we define a compressed matrix $\tilde{D}$, whose first entry is $\frac{1}{w_1} \sum_{l=1}^{w_1} D_l$, and so on. Then we have $\hat{B} = (\tilde{Q}'W\tilde{Q})^{-1}\tilde{Q}W\tilde{D}$, where $W = \text{diag}(w_1, \dots, w_{n^*})$ is the weight matrix. This $\hat{B}$ is exactly the same as in (6) and can be used for computational convenience.

Here we list three main conditions and two theorems for $\hat{B}$, the proof can be found in the appendix.

(A.1) $|X_i| \leq M < \infty$ a.s.

(A.2) The random vector $(Z_{q,2}, \dots, Z_{q,p}, X_i^T Z_{q,2}, \dots, X_i^T Z_{q,p})$ has non-singular covariance matrix Σ

(A.3) $\sup_{u,v} |f(u) - f(v)| \leq L|u - v|$ and $\sup_{u,v} |g_i(u) - g_i(v)| \leq L|u - v|$ a.s.

Theorem 1. Suppose A.1-A.3 hold, $m, n \rightarrow \infty$, and $\sqrt{n}/m \rightarrow 0$. The estimator $\hat{B}$ given in (6) is consistent and for $m^* = \lfloor \frac{m^2}{2} \rfloor$, $\sqrt{nm^*}(\hat{B} - B) \xrightarrow{L} N(0, \sigma^2 \Sigma^{-1})$.

Theorem 2. Under A.1-A.3, for testing $H_0 : CB = 0$ against $H_1 : CB \neq 0$, where C is an $s \times p$ with $\text{rank}(C) = s$, the test statistic

$$F = \frac{\hat{B}C'(C(Q'Q)^{-1}C')^{-1}C\hat{B}/s}{\hat{\sigma}^2}$$

asymptotically follow the $F(s, nm^* - p)$ distribution under the null hypothesis, with $m^* = \lfloor \frac{m^2}{2} \rfloor$.

Algorithm Part II: If other parts of the model are of interest, we could further obtain the estimates of α_1 , β_1 , f and g_i through nonparametric smoothing steps.

1. Let

$$y'_q = \frac{1}{n} \sum_{i=1}^n [y_{i,q} - \sum_{p=2}^{r^2} (\hat{\alpha}_p Z_{q,p} + \hat{\beta}_p X_i \times Z_{q,p})].$$

Then $\hat{f}(\cdot)$ can be obtained by a suitable nonparametric smoothing method using data $\{(d_q, y'_q), q = 1, \dots, m^2\}$. We adopt a local linear smoother in this paper (Fan & Gijbels 1996). For any d , minimizing

$$\sum_{q=1}^{m^2} \mathcal{K}\left(\frac{d - d_q}{h}\right) (y'_q - b_0 - b_1(d - d_q))^2$$

with respect to b_0 and b_1 , leads to $\hat{f}(d) = \hat{b}_0$, where h is the bandwidth and $\mathcal{K}$ is a kernel function. For identifiability, we let $\hat{\alpha}_1 = \hat{f}(0)$ and $\hat{f}(\cdot) = \hat{f}(\cdot) - \hat{\alpha}_1$.

2. Let

$$y_i'' = \frac{1}{m^2} \sum_{q=1}^{m^2} [y_{i,q} - \sum_{p=2}^{r^2} (\hat{\alpha}_p Z_{q,p} + \hat{\beta}_p X_i \times Z_{q,p}) - \hat{\alpha}_1 - \hat{f}(d_q)].$$

$\hat{\beta}_1$ is obtained from a simple linear regression on $\{X_i, y_i''\}$.

3. Let $\tilde{y}_{i,q} = y_{i,q} - \sum_{p=1}^{r^2} (\hat{\alpha}_p Z_{q,p} + \hat{\beta}_p X_i \times Z_{q,p}) - \hat{f}(d_q)$. Then the local linear estimator of $\hat{g}_i(d)$ can be obtained by minimizing

$$\sum_{q=1}^{m^2} \mathcal{K}\left(\frac{d - d_q}{h}\right) (\tilde{y}_{i,q} - b_0 - b_1(d - d_q))^2$$

with respect to b_0 and b_1 , and $\hat{g}_i(d) = \hat{b}_0$, where h is the bandwidth and $\mathcal{K}$ is a kernel function.

2.3 Data Driven Brain Partition

For brain partition, we could perform individual partition which leads to different labeling for each subject, or we could perform partition on the group average data which leads to the same brain partition for all subjects. We note here that our proposed modeling in Section 2.1 works for both cases, although we omit the subject index i in the dummy variables Z for notation simplicity. Now let's take one subject as an example to describe the data driven brain partition procedure. Let $\tilde{y}_{jl} = y_{i,jl}$, $C = C(X_i)$ and $L(d_{jl}) = f(d_{jl}) + g_i(d_{jl})$. We estimate labels $\mathbf{R}$, by minimizing the following least square modularity

$$\min_{\mathbf{R}} \min_{L,C} \sum_{j,l=1}^m (\tilde{y}_{jl} - R_j' C R_l - L(d_{jl}))^2. \quad (7)$$

We need to minimize (7) over all possible partitions $\mathbf{R}$. The problem is NP-hard and we used a modified labeling switch algorithm to approximate the optimal solution (Stephens

2000; Bickel & Chen 2009). Starting with some initial labeling values $\mathbf{R}$ (for example, randomly assign a value from $\{1, \dots, r\}$ to each node), iterate the following steps until a fixed point is reached or the total sum of residual squares does not decrease any more.

1. For a given label matrix $\mathbf{R}$, estimate $\hat{C}$ and $\hat{L}$ as described below.
2. For each $j = 1, \dots, m$, update R_j by finding $k^* \in \{1, \dots, r\}$ and R_j equals 1 at entry k^* , to minimize

$$\sum_{l=1}^m (\tilde{y}_{jl} - R'_j \hat{C} R_l - \hat{L}(d_{jl}))^2.$$

For a fixed label matrix $\mathbf{R}$, we can estimate L and C through a backfitting algorithm (Hastie and Tibshirani (1986)). Each iteration contains the following two steps:

1. Given the estimates of $\hat{L}^t$ at the t -th step, for all $k, k' = 1, \dots, r$,

$$\hat{C}_{k,k'}^{t+1} = \frac{1}{m_k \times m'_{k'}} \sum_{\{j \text{ in region } k\}} \sum_{\{l \text{ in region } k'\}} [\tilde{y}_{jl} - \hat{L}(d_{jl})],$$

where m_k is the number of nodes in community k .

2. Given the estimates of $\hat{C}^{t+1}$, we propose to estimate L^{t+1} through nonparametric smoothing on the data $\{d_{jl}, \tilde{y}_{jl} - \hat{C}_{R_j R_l}^{t+1}\}$, such as local linear smoothing, smoothing spline and p-spline methods. We used smoothing spline with Matlab function *fit*.

3 Data Application

We apply the proposed method to a resting-state functional magnetic resonance imaging (RS-fMRI) data set to study the development of the basal ganglia circuitry. One hundred and thirty-nine participants aged 10-20 years participated in the study and RS-fMRI data were acquired for each participant at the University of Pittsburgh Medical Center. Forty participants were excluded due to excessive head motion, poor coverage in the cerebellum, or

peripheral physiological equipment failure. Imaging data were preprocessed using AFNI (Cox et al. 1996) and FSL (Smith et al. 2004). Anatomic scans were simultaneously segmented into white matter, gray matter, ventricle, and non-brain tissue voxels, and normalized to the MNI152 template using FSLs non-linear registration procedure. Detailed preprocessing steps can be found in Hwang et al. (2012), including slice-timing correction, motion correction, co-registration and de-trending. We conduct our analyses on $n = 97$ subjects and $m = 448$ gray matter voxels in the basal ganglia mask.

RS-fMRI measures intrinsic, high-amplitude, low-frequency blood-oxygen-level dependence signal (BOLD) fluctuations of the brain and correlated RS-fMRI signal reflects functional connectivity independent of any particular brain state (Van Dijk et al. 2010). The basal ganglia subserve a wide range of functions, including motor, cognitive, motivational, and emotional processes and has been implicated in numerous neurological and psychiatric disorders. There have been great interest in using RS-fMRI techniques to study the functional connectivity in basal ganglia (Di Martino et al. 2008; Robinson et al. 2009; Barnes et al. 2010).

First we apply the data driven brain partition on the group averaged connectivity matrix $(y_{:,j})$, which contains the Fisher’s z-transformed correlation values of resting-state BOLD signal fluctuations among voxels. The five functional distinct brain regions are visualized in the top-left panel of Figure 1 and also some snapshots of the parcellation are given in Figure 2. Comparing the data driven partition with anatomy, we find that cluster 1 should be caudate body (dark blue in Figure 1), cluster 3 is putamen (green), cluster 5 is pallidum (dark red) and cluster 2, 4 could be caudate head, but they also spread out to places outside the caudate. Clusters 1, 3, 5 are clean and seem to match well with the anatomy. Cluster 2 and 4 could still be interesting if they reflect common input/output of the basal ganglia circuitry, which needs further investigation. These findings are very interesting and it is purely based on data driven statistical methods. We note that further studies on a larger

mask and even whole brain data is our next step application. Based on this partition, using methods proposed in section 2.2, we can estimate all the components in model (1). The estimated f function is presented in the bottom of Figure 1 in two ways. Estimated g_i functions for four randomly selected subjects are shown in Figure 3. The top-right panel of Figure 1 show the group averaged data $y_{.,jl}$ and it clearly matches our semiparametric blockmodel $R'_j C R_l + f(d_{jl})$.

The main inference of interest is how the functional connectivity among these functionally distinct regions 1-5 develops with age and interacts with cognitive measures such as IQ. According to previous studies, differential patterns of connectivity would be noted across the subregions examined. More specifically, we would like to test if the development of functional connectivity in the basal ganglia shows a “local to distributed” pattern (Fair et al. 2009). Here we define the contrasts between inter-regional (distributed) connectivity and intra-regional (local) connectivity as follows.

For any $1 \leq k < k' \leq r$,

$$\Delta_{kk'}(X) = 2C_{kk'}(X) - C_{kk}(X) - C_{k'k'}(X).$$

We note that $C_{11}(X) = \alpha_{11} + \beta_{11}X$ and other than $k = k' = 1$, we have $C_{kk'} = (\alpha_{11} + \alpha_{kk'}) + (\beta_{11} + \beta_{kk'}X)$. Therefore

$$\Delta_{kk'}(X) = \begin{cases} (2\alpha_{kk'} - \alpha_{kk} - \alpha_{k'k'}) + (2\beta_{kk'} - \beta_{kk} - \beta_{k'k'})X & \text{for } 1 < k < k' \leq r \\ (2\alpha_{kk'} - \alpha_{k'k'}) - (2\beta_{kk'} - \beta_{k'k'})X & \text{for } 1 = k < k' \leq r. \end{cases}$$

Define

$$\Delta\beta_{kk'} = \begin{cases} 2\beta_{kk'} - \beta_{kk} - \beta_{k'k'} & \text{for } 1 < k < k' \leq r \\ 2\beta_{kk'} - \beta_{k'k'} & \text{for } 1 = k < k' \leq r. \end{cases}$$

Here we consider two covariates *age* and *IQ*, so $\Delta\beta_{kk'} = (\Delta\beta_{kk,1}, \Delta\beta_{kk,2})$ for *age* and *IQ* respectively, and we are interest in testing the null hypothesis that

$$\Delta\beta_{kk',d} = 0$$

for $1 \leq k < k' \leq r$ and $d = 1, 2$.

The hypothesis test results are shown in Table 1. Under the Bonferroni-corrected level 0.005, we find *age* has significant positive effect on four pairs of regions $2 \leftrightarrow 3$, $2 \leftrightarrow 5$, $3 \leftrightarrow 4$ and $4 \leftrightarrow 5$, which means the connectivity between these regions show a “local-to-distributed” pattern as age increases. We also find *IQ* is positively correlated with the “local-to-distributed” pattern in regions $1 \leftrightarrow 2$ and $2 \leftrightarrow 5$.

4 Numerical Study

We conducted two numerical studies. The first was designed to evaluate the proposed method when sample size n , number of nodes m , and signal to noise ratios are changing. We report the estimation errors for all α , β , σ^2 , f and g_i in Table 2 and Figure 4. We also report the power of the proposed test under different scenarios in Table 3. In the second study, we generated data sets that closely resemble the RS-fMRI data described in Section 3. We can see from Table 4 that the proposed method has good power where true signals are present, while type 1 error rates are well controlled.

In the following we discuss the details of the simulation. We ran 200 simulations for each numerical study, and data $y_{i,jl}$ were generated from Eq. (4), i.e.,

$$y_{i,jl} = \sum_{1 \leq k \leq k' \leq r} (\alpha_{kk'} Z_{jl,kk'} + \beta_{kk'} X_i \times Z_{jl,kk'}) + f(d_{jl}) + g_i(d_{jl}) + \epsilon_{ijl}.$$

In the first study, we generated $X_i \sim N(0, 1)$, $i = 1, \dots, n$, and three-dimensional coordinates of node j , $\{\omega_j = (\omega_{j,1}, \omega_{j,2}, \omega_{j,3}), j = 1, \dots, m\}$, were generated independently from uniform $[0, 10]$. We compute $d_{jl} = \|\omega_j - \omega_l\|_2$. Here $f_d = d + \sin(d)$, $g_i(d) = \sum_{v=1}^5 \xi_{iv} \phi_v(d)$, $\phi_v(d) = \cos((v+1)\pi d/10)/\sqrt{5}$ for odd values of v , and $\phi_v(d) = \sin(v\pi d/10)/\sqrt{5}$ for even values of v . The ξ s were generated as $\xi_{iv} \stackrel{iid}{\sim} N(0, \lambda_v)$, and the corresponding $\lambda = \{25/55, 16/55, 9/55, 4/55, 1/55\}$. The measurement errors $\epsilon_{i,jl}$ are i.i.d. $N(0, 1)$. We tried

$n = \{30, 60, 100\}$ and $m = \{100, 400\}$, and all $\alpha_{kk'}$ and $\beta_{kk'}$, $k, k' = 1, 2$ were set to be zero except that β_{12} was one of $\{0, 0.5, 1\}$.

For those coefficients and $\hat{\sigma}^2$ estimated from the difference based procedure in steps 1-3, the errors are very small even in the case of $n = 30$ and $m = 100$, and get better as n, m increase. The estimation of f and g are quite good. The true and estimated functions from one simulation run are illustrated in Figure 3. The estimates of α_{11} and β_{11} are also quite good, though not as accurate as other coefficients since the error of these two depends on $\hat{f}$ and $\hat{g}$. We also report the power for testing $\beta_{12} = 0$ in Table 3, which demonstrates the great performance of the proposed procedure.

In the second simulation, we set $n = 97$, $m = 448$, $X_i = (age_i, IQ_i)^T$ for $i = 1, \dots, n$. We used the same d_{jl} , $j, l = 1, \dots, m$ as in the real data set. The functions f were estimated from the data set, as shown in Figure 1. In each simulation, we bootstrapped g_i , $i = 1, \dots, n$ from the sets of random functions g_i estimated from the data as shown in Figure 3. The variance of the measurement errors was estimated to be $\hat{\sigma}_\epsilon^2 = 0.329$, and then $\epsilon_{i,jl} \stackrel{iid}{\sim} N(0, \hat{\sigma}_\epsilon^2)$. For the coefficients, we set $\alpha_{kk'} = \hat{\alpha}_{kk'}$ and $\beta_{kk'} = 0$ for $k, k' = 1, \dots, 5$, except those found to be significant in the data analysis. To be more specific, we set $\Delta\beta_{12} = \beta_{12} = (0, 0.0019)$, $\Delta\beta_{23} = \beta_{23} = (0.0047, 0)$, $\Delta\beta_{25} = \beta_{25} = (0.0064, 0.0025)$, $\Delta\beta_{34} = \beta_{34} = (0.0030, 0)$, and $\Delta\beta_{45} = \beta_{45} = (0.0038, 0)$, where the first number corresponds to the effect of *age*, and the second number corresponds to the effect of *IQ*, and all values are estimated from the RS-fMRI data set.

As mentioned earlier, the proposed model and inference starts with a partition of the nodes. Here we first investigated the performance of the testing procedure for a provided brain partition. We report the results in the top two rows of Table 3, which show that the proposed method has high power when true signal is present and meanwhile successfully controls the type I error. To investigate the effect of the estimation error in a possible data-driven partition step, we applied the partition algorithm as discussed in Section 2.3 for each

simulated data and estimated the model components and perform hypothesis testing based on the estimated labels R . Simulation results are shown in the bottom two rows of Table 3. It demonstrates that the proposed inference is not sensitive to moderate errors in the partition step.

5 Discussion

Most of the current methods for network inference are two-step procedures where network modeling and inference of covariate effects are conducted in separate steps. We believe that this work is an important step toward model based inference for network data. The proposed model decomposes the connectivity matrix $y_{i,jl}$ into a low rank block structure $R_j' C(X_i) R_l$ and a nonparametric smooth part $f(d_{jl}) + g_i(d_{jl})$. Here two types of covariates are incorporated in the model. One is the node level covariates d_{jl} . In our application, d_{jl} is the Euclidean distance between spatial locations of voxel j and voxel l . In other applications, this can be a distance in abstract sense that measures some type of difference between two nodes, such as age difference in social network. Or more generally, d_{jl} could be more than one dimension measuring several distances in different aspects, in which case, ideas from single index model can be borrowed and the nonparametric part can be modeled as $f(\theta^T d_{jl})$, with $\|\theta\|_2 = 1$ for identifiability. A second type of covariates considered is the subject level covariate X_i . In our application, we are mainly interested in the effects of X_i on the block connectivity matrix C . This is a parsimonious model to reduce large number of voxel level inferences to relatively small number of regional level inferences. Several variations of blockmodel such as degree corrected blockmodel might be considered to gain more flexibility in appropriate applications.

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Appendix: proofs

Proof of Theorem 1:

Note that $\sqrt{nm^*}(\hat{B}-B) = \sqrt{nm^*}(Q'Q)^{-1}Q(\delta+\varepsilon)$, and $\varepsilon \sim N(0, \sigma^2 I_{nm^*})$. $\sqrt{nm^*}(Q'Q)^{-1}Q\varepsilon \xrightarrow{L} N(0, \sigma^2 \Sigma^{-1})$ follows from the standard central limit theorem. So we only need to show that $\sqrt{nm^*}(Q'Q)^{-1}Q\delta \xrightarrow{P} 0$. We note that $\frac{1}{nm^*}(Q'Q) \rightarrow E(Q'Q) \rightarrow \Sigma$, and $Q\delta = \sum_{i=1}^n \sum_{q=1}^{m^*} Q_{i,q} \delta_{i,q} \leq 2nC \sum_{q=1}^{m^*} |d_{2q} - d_{2q-1}|L$, a.s., where C is a constant that bounds the maximum absolute value of $Q_{i,q}$ (using A.1), and L is the Lipschitz constant (using A.3). Since d_q are ordered from smallest to largest, $\sum_{q=1}^{m^*} |d_{2q} - d_{2q-1}|$ is bounded by the maximum Euclidean distance between two nodes, which is a bounded constant since we are considering nodes embedded in a fixed domain. Therefore we have $\sqrt{nm^*}(Q'Q)^{-1}Q\delta = O_p(n/\sqrt{nm^*}) = O_p(\sqrt{n}/m) \xrightarrow{P} 0$.

Proof of Theorem 2:

It follows from Theorem 1 that if σ^2 is given, $\frac{\hat{B}C'(C(Q'Q)^{-1}C')^{-1}C\hat{B}/s}{\sigma^2}$ has a limiting χ^2 distribution with s degrees of freedom. To estimate σ^2 , let $H = I - Q(Q'Q)^{-1}Q'$, we have that $\hat{\sigma}^2 = \frac{\|HD\|_2^2}{nm^*-p}$, where we assume Q is full rank. Note that $\|HD\|_2^2 = \epsilon'H\epsilon + 2\epsilon'H\delta + \delta'H\delta$. It is easy to see that $\delta'H\delta \xrightarrow{a.s.} 0$ as $n \rightarrow 0$ (using A.3). Since $\epsilon'H\delta|\delta \sim N(0, 2\sigma^2\delta'H\delta)$, we know that $\epsilon'H\delta \xrightarrow{a.s.} 0$. Following standard arguments, it is easy to show that $\sigma^{-2}\epsilon'H\epsilon$ is $\chi_{nm^*-p}^2$. This completes the proof.

Table 1: P-values for testing $\Delta\beta_{jl} = 0$ VS $\Delta\beta_{jl} \neq 0$. We bold those P-values that are less than 0.005, and we note that the estimated values are all positive for those bolded pairs. We found that the connections are more distributed in older subjects for regions $2 \leftrightarrow 3$, $2 \leftrightarrow 5$, $3 \leftrightarrow 4$ and $4 \leftrightarrow 5$. And also the distributed connections are positively associated with IQ values in pairs of regions, $1 \leftrightarrow 2$, and $2 \leftrightarrow 5$.

	$\Delta\beta_{12}$	$\Delta\beta_{13}$	$\Delta\beta_{14}$	$\Delta\beta_{15}$	$\Delta\beta_{23}$	$\Delta\beta_{24}$	$\Delta\beta_{25}$	$\Delta\beta_{34}$	$\Delta\beta_{35}$	$\Delta\beta_{45}$
age	0.0056	0.4825	0.7180	0.2908	0.0000	0.9007	0.0000	0.0017	0.0457	0.0002
IQ	0.0000	0.5341	0.3368	0.2792	0.0062	0.0302	0.0000	0.8592	0.9074	0.0333

Table 2: Results for *Simulation 1*: Average Mean Squared Errors (over 200 simulation runs) for the estimated coefficients and error variance for various sample sizes n and numbers of nodes m . The true values: $\beta_{12} = 0.5$, $\sigma^2 = 1$, and zero otherwise.

	m = 100			m = 400		
	n=30	n=60	n=100	n=30	n = 60	n=100
$MSE(\hat{\alpha}_{12})$	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000
$MSE(\hat{\alpha}_{22})$	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000
$MSE(\hat{\beta}_{12})$	0.0001	0.0001	0.0000	0.0000	0.0000	0.0000
$MSE(\hat{\beta}_{22})$	0.0001	0.0001	0.0000	0.0000	0.0000	0.0000
$MSE(\hat{\sigma}^2)$	0.0004	0.0004	0.0003	0.0002	0.0002	0.0002
$MSE(\hat{\alpha}_{11})$	0.0028	0.0020	0.0010	0.0022	0.0014	0.0009
$MSE(\hat{\beta}_{11})$	0.0008	0.0005	0.0002	0.0004	0.0002	0.0001

Table 3: Results for *Simulation 1*: Power (over 200 simulation runs) for testing $\beta_{12} = 0$ VS $\beta_{12} \neq 0$, for various sample sizes n and numbers of nodes m . We control type I error at 0.05.

β_{12}	m = 100			m = 400		
	n=30	n=60	n=100	n=30	n = 60	n=100
0	0.04	0.04	0.06	0.04	0.04	0.05
0.5	1.00	1.00	1.00	1.00	1.00	1.00
1	1.00	1.00	1.00	1.00	1.00	1.00

Table 4: Results for *simulation 2*, where the simulated data closely resembles the real data set as discussed in Section 3: Power (over 200 simulation runs) for testing $\Delta\beta_{jl} = 0$ VS $\Delta\beta_{jl} \neq 0$, for a fixed partition (top two rows) and data-driven partitions (bottom two rows). We control the family wise type I error rate at 0.05 level and Bonferroni correction was used to adjust the individual test level. We bold the numbers where the true signals are present.

	$\Delta\beta_{12}$	$\Delta\beta_{13}$	$\Delta\beta_{14}$	$\Delta\beta_{15}$	$\Delta\beta_{23}$	$\Delta\beta_{24}$	$\Delta\beta_{25}$	$\Delta\beta_{34}$	$\Delta\beta_{35}$	$\Delta\beta_{45}$
age	0.0000	0.0100	0.0100	0.0100	1.0000	0.0000	1.0000	1.0000	0.0050	1.0000
IQ	1.0000	0.0000	0.0050	0.0000	0.0100	0.0000	1.0000	0.0000	0.0000	0.0100
age	0.0050	0.0050	0.0000	0.0200	0.9850	0.0100	0.9850	0.9950	0.0050	0.9850
IQ	0.9850	0.0001	0.0002	0.0000	0.0000	0.0000	0.9950	0.0150	0.0200	0.0200

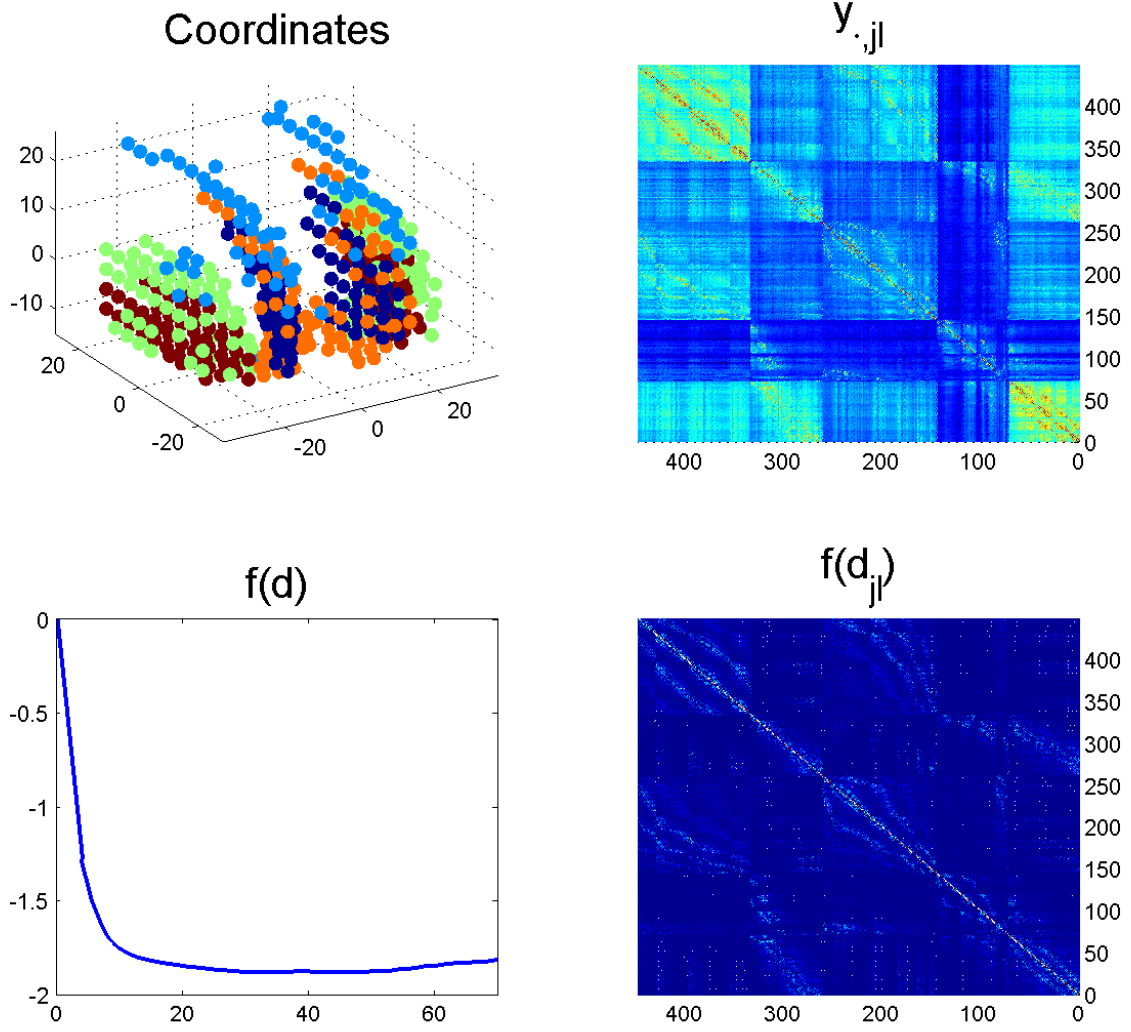


Figure 1: Figures for the RS-FMRI data. Top left: the embedding of the 448 nodes in the three dimensional brain space. Each color denotes a functionally distinct region. Top right: the average (over 97 subjects) connectivity measure $y_{.,jl}$ for $j, l = 1, \dots, 448$ (448 voxels are ordered by 5 regions). Bottom left: the estimated function $f(d)$. Bottom right: the estimated f function evaluated at d_{jl} . One can clearly see patterns that $y_{.,jl} \sim R_j' C R_l + f(d_{jl})$.

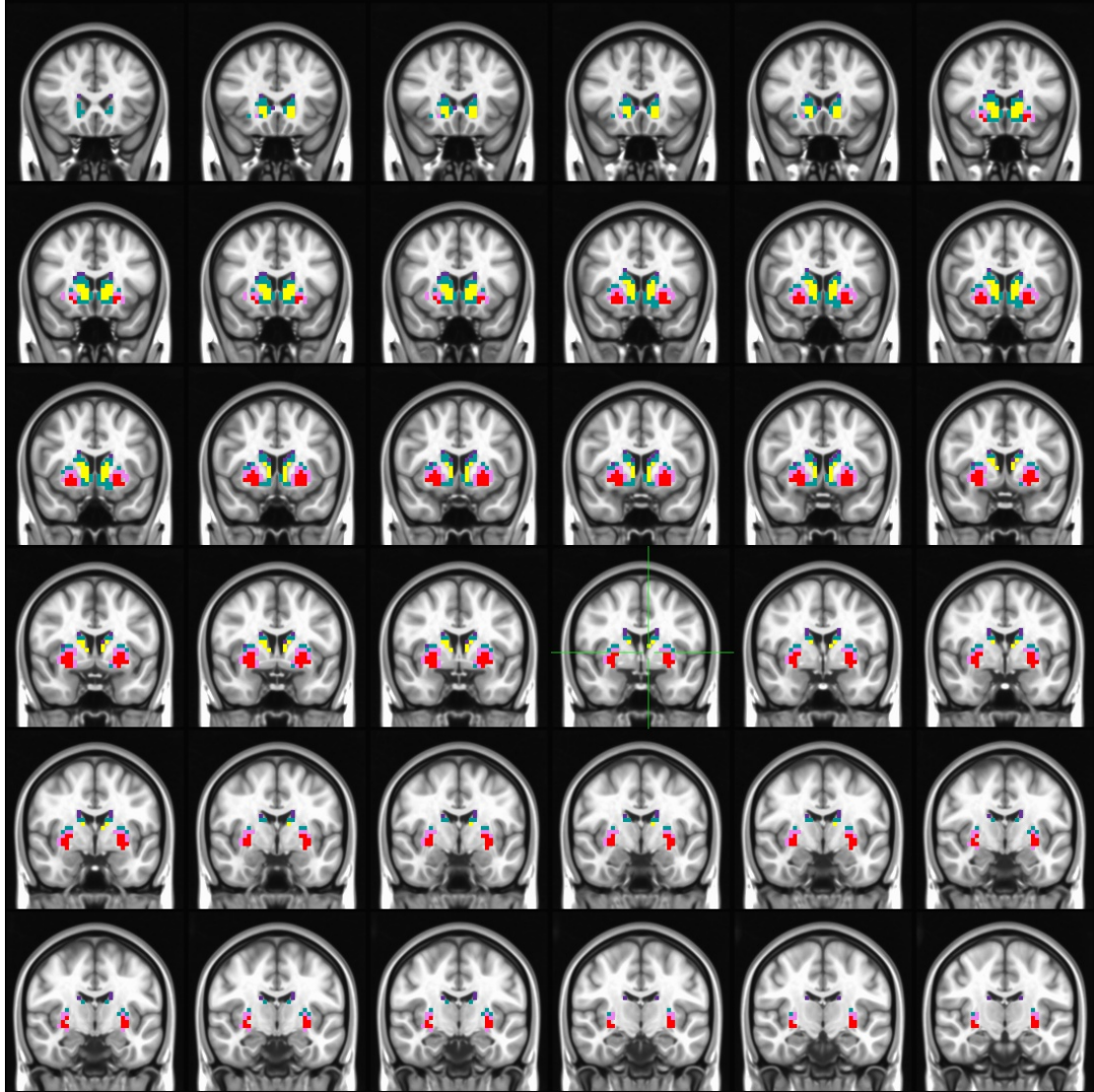


Figure 2: Some snapshots of the data-driven brain partition for the basal ganglia mask using the RS-fMRI data.

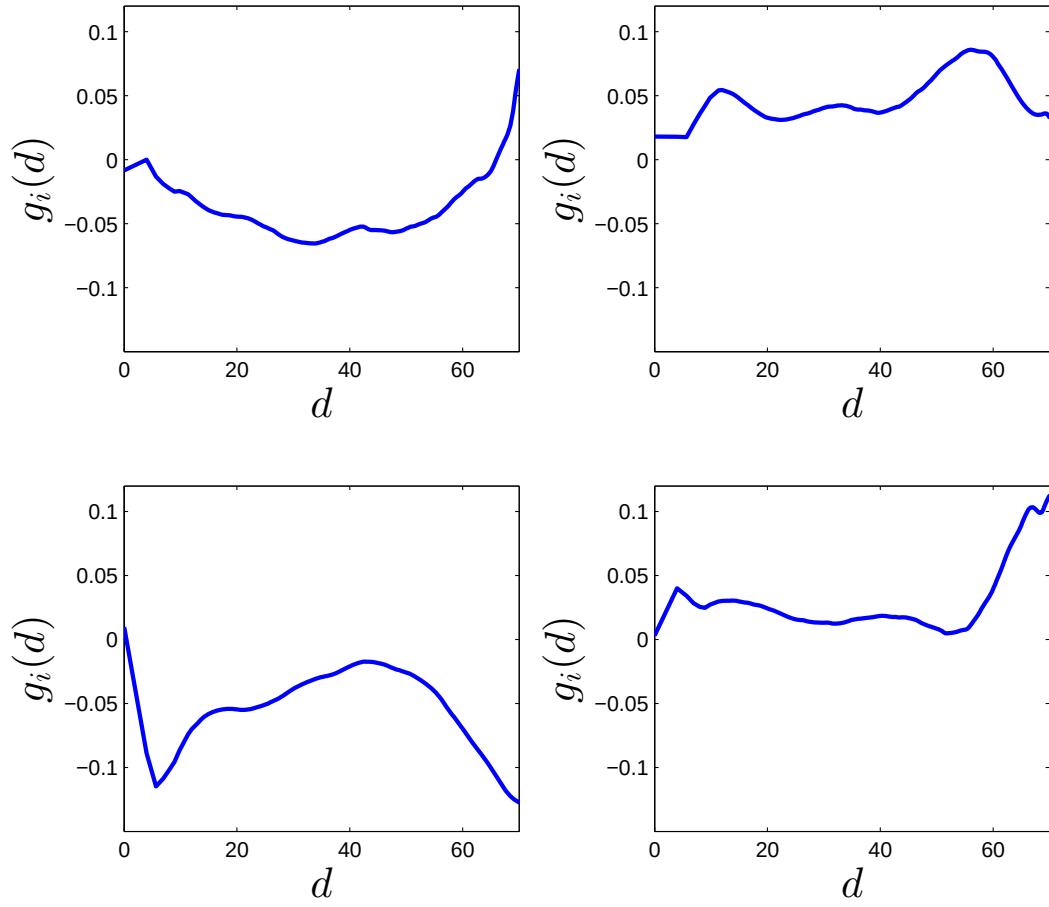


Figure 3: For the RS-fMRI data application, the estimated g_i for four randomly selected subjects.

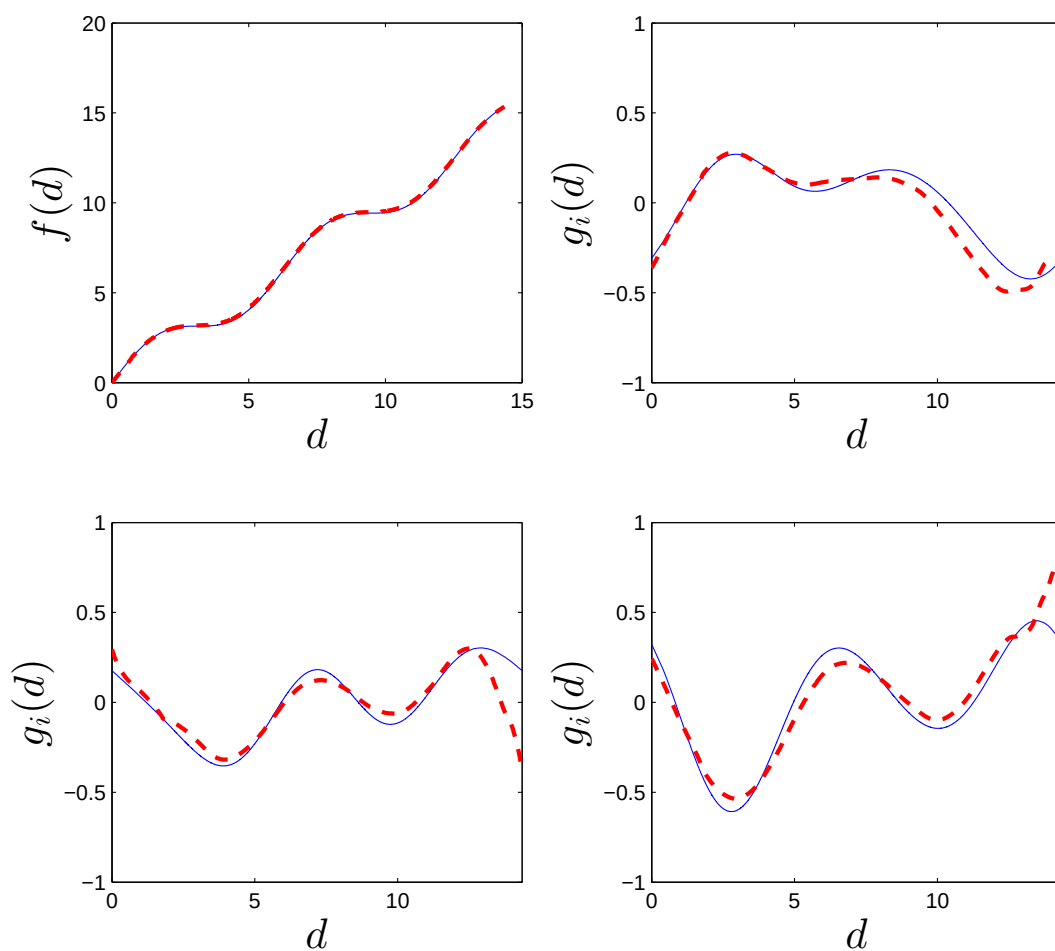


Figure 4: For *simulation 1*, the true (blue solid lines) and estimated (red dashed lines) functions of f (top left) and three randomly selected g_i .